

The Fundamentals of Programming

Lists



Objectives

What you will learn:

1. How lists in Python works.
2. How to add, remove and access items within a list.
3. To loop over lists using both for loops and foreach loops.

List

The basics

Much of what you will learn about lists in Python is going to remind you of Strings.

Take a look at the code to the right, and using what you know about Strings and functions, try and determine the answer.

```
lst = [1, 2, 3, 4, 5, 6]
```

```
print(lst[0])  
print(lst[4])  
print(lst[1:4])  
print(len(lst))  
print(lst[1] + lst[5])  
print(lst[10])
```

Using Python Lists

With lists you have access to many helpful functions that allow you to add, remove and access elements and indices.

```
alpha = ['a', 'b', 'c', 'd', 'e']
```

Append:

```
alpha.append('f')
```

Remove:

```
alpha.remove('c')
```

Pop:

```
alpha.pop(3)
```

Index:

```
alpha.index('b')
```

List functions

Practice

For each of the examples use the **nums** list and find the output. You can do a search for the function if you can't deduce what it does.

nums = [1, 3, 5, 2, 0]

```
var = len(nums)
nums.append(var)
print(nums)
```

```
nums.pop(3)
nums.pop(2)
nums.append(6)
print(nums)
```

```
nums.remove(5)
nums.pop(2)
nums.remove(1)
nums.pop(2)
print(nums)
```

Loops and lists

Practice

For loops make working with list incredibly easy. With the following problems use the list below and state the output.

`lst = [1, 2, 3, 5, 7, 8, 9]`

```
for i in range(len(lst)):  
    num = lst[i] * 2  
    print(num)
```

How could we simplify this?

```
for i in range(len(lst)):  
    lst[i] = lst[i] % 2  
    print(lst)
```

What does the list tell us about the original numbers?

The foreach loop

Sometimes you don't want to use the indices with the list and you just want the elements inside. That's what the foreach loop is for.

```
alpha = ['a', 'b', 'c', 'd', 'e']
```

With a regular for loop:

```
for i in range(len(alpha)):
    print(alpha[i])
```

The for each loop:

```
for item in alpha:
    print(item)
```

When would want to use each of these loops?

Which loop to use?

Practice

With each of these problems figure out which type of loop would be best and then solve it.

Given a list of numbers called `nums = [1, 2, 3, 4]` add 3 to each of the elements so that the list becomes: `[4, 5, 6, 7]`

Given the list: `grades = [54, 78, 94, 91, 44]` print all the grades to the screen, one grade per line and as a floating point number (ex. 54.0)

Recap

We learned

1. How to create a list.
2. How to **append**, **remove**, and **pop** elements from a list.
3. How to find the **index** of an item in a list.
4. To find the length of a list using **len**.
5. The difference between for and foreach loops.